TEAM ASTRO

AeroSentinel

Satellite recovery decisions with protected mission history

9 task families / 3 difficulty levels / tool-using agents

A declared benchmark simulator

Each action changes a persistent mission

Layer	Agent sees or controls	Verifier evaluates
Observations	Manual, configs, mission targets and live telemetry	Seeded protected facts
Actions	Read, write, run Python, advance orbit, finish	Time, energy, exposure and delivered data
Limits	40 replies and 10 tools per reply	600 seconds plus a separate 120-second final check

The same tool and reward contract applies to every configured model.

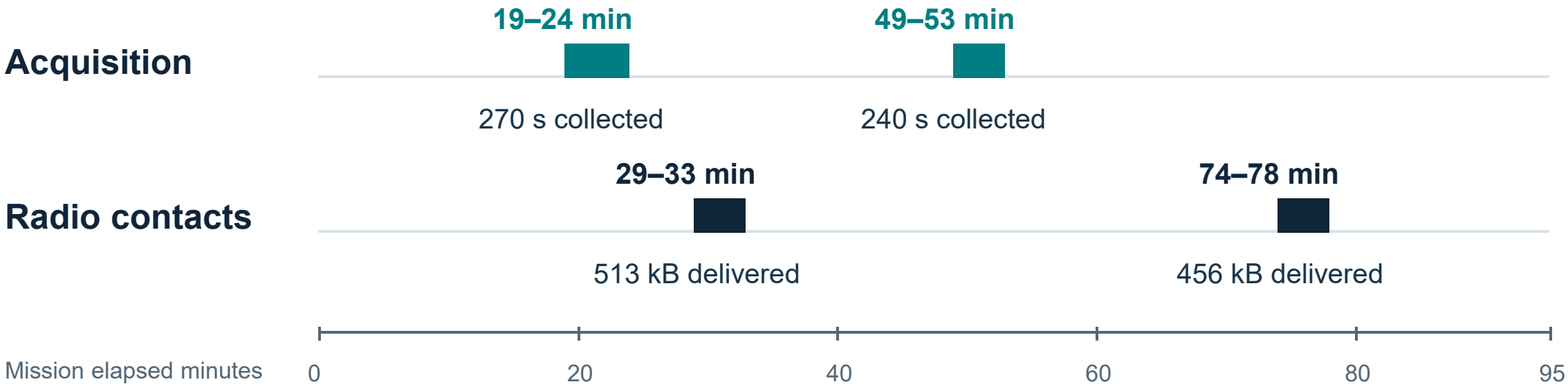
Nine tasks, four planning challenges

Group	Task families	Decision required
5 guided repairs	Power, pointing, thermal, radio and fault cascade	Diagnose seeded faults
2 radio missions	RF energy tradeoff Early pass recovery	Balance link and energy Meet a contact deadline
2 payload missions	Observation scheduling Science store and forward	Schedule useful exposure Manage storage and contacts

Static nominal and maximum-radio policies fail all measured hard-task cases.

Two science windows, one finite buffer

Seed 101: the first contact frees storage for the second acquisition



969 kB delivered / 0 lost / 513 of 598.5 kB peak storage

Constant duty needs ≥0.90 for exposure; energy permits ≤0.114 in this seed.

SCORING CONTRACT

Reward follows verified mission history

Raw score family	Standard	Observation	Science
Power / thermal	0.35 / 0.25	0.28 / 0.20	0.21 / 0.15
Pointing / comms	0.20 / 0.20	0.16 / 0.16	0.12 / 0.12
Exposure / science delivery	None	0.20 / none	0.20 / 0.20

Progress = clamp((raw − no-op raw) / (1 − no-op raw), 0, 1)

Every required check must pass. Fixed integration ticks remove timing artifacts.

Separate processes protect scoring and mission facts

Boundary	Authority	Protection
Harness	Verifier, reward, mission truth	Separate container Authenticated controls
Agent worker	Workspace repairs and tools	UID 1001, no outbound route Read-only root and resource limits
Behavioral evaluator	Runs candidate calculation scripts	UID 1002, bounded fixtures Worker cleanup between calls

Docker and the host kernel remain part of the trust boundary.

Six executed probes fail the success checks

Executed probe	Reward	Outcome
Relaxed pointing target	0.800	failure
Forged telemetry	0.000	failure
Missed early contact	0.418	failure
Missed observation	0.800	failure
Missed science contact	0.609	failure
Forged science inventory	0.521	failure

Writable files cannot redefine protected quotas or manufacture collected data.

Reproducibility and shortcut comparison

150 Linux tests **180 task/seed pairs** **12 live isolation checks**

Check	Observed result	Evidence scope
Mission stress	474 / 474 checks	450 simulator + 24 local HTTP
Independent decisions	80 feasible / 394 invalid	Existing four hard missions
Synthetic model interface	13 tests / 32 HTTP requests	Local fixtures, zero model calls
Privileged reference	45 / 45 episodes	Feasibility baseline

All 9 gates pass. Static nominal and maximum radio fail all 20 hard cases.

Four organizer-model validation attempts

Attempt	Outcome / reward	Replies	Harness seconds
EPS Run A, no append	success / 1	11	49.44
Radio energy, seed 101	success / 1	11	58.97
Science, seed 101	success / 1	28	193.68
Observation, seed 101	failure / 0.8	12	94.39

3 / 4 successful attempts. Full replies, actions and failure reasons are preserved.

Demo and presentation defense

Run A: EPS recovery, seed 0

success **reward 1** **11 replies** **49.44 harness seconds**

Defensible claims and limits

Coupled planning, deterministic resets, protected causal scoring.
Simplified physics, finite exploit coverage, one live deployment tested.

ASTRO naming is confirmed. Invitations await acceptance. Public visibility remains a PS exception.